

Generation Interconnection Request System Impact Study Report

For Queue Project Q14-0343

Estill Springs 100 kV
July 26, 2015

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q14-0343. The Interconnection Customer proposes a 24.0 MW generating facility interconnecting at the Estill Springs 100 kV substation in the Illinois area. The request was received on June 17, 2014. The study evaluated system intact and single contingency conditions using the regional planning case, including 220 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$138,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: ORCHARD ROAD – FORT HUMBUG 500kV
Contingency: FORT HUMBUG – WEST CAMPUS 345kV
Loading: 105.0% (Rating: 3,014 MVA)
DFAX: 12.6%
Pre-project loading: 104.9%
2. Facility: W CAMPUS – FT HUMBUG 345kV
Contingency: FORT HUMBUG – ORCHARD RD 500kV
Loading: 103.7% (Rating: 2,193 MVA)
DFAX: 8.4%
Pre-project loading: 103.6%
3. Facility: LITTLE CHUTE – TALENENERGY MARTINS CREEK LLC HARRISBURG 345kV
Contingency: FT HUMBUG SUBSTATION – CORONET 500kV
Loading: 102.6% (Rating: 1,626 MVA)
DFAX: 7.0%
Pre-project loading: 102.5%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
FT HUMBUG – ORCHARD RD 500kV line	Reconductor line	\$34,161,000	0% (\$81,000)
W CAMPUS – FORT HUMBUG 345kV L	Reconductor line	\$18,395,000	0% (\$43,000)
TALENERGY MARTINS CREEK LLC HARRISBURG – LITTLE CHUTE 345kV	Reconductor line	\$5,929,000	0% (\$14,000)
	Total allocated to project		\$138,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.